**King Fahd University of Petroleum and Minerals**
**Department of Mathematics**
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# Math 101 Final Exam Term 242

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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**1.** The derivative of the function

$$f(x) = \ln \sqrt{\frac{x+1}{x-1}} + \frac{\cos x}{\csc x}$$

is

(a) $\dfrac{x-1}{x+1} - 1$

(b) $\dfrac{1}{1-x^2} + \cos 2x$

(c) $\dfrac{1}{1-x^2} + \dfrac{-\sin x - \cot x \cos x}{\csc x}$

(d) $\dfrac{x-1}{x+1} + \cos 2x$

(e) $\dfrac{x-1}{x+1} + 1$

**2.** The minimum vertical distance between the parabolas $y = x^2 + 1$ and $y = x - x^2$ is

(a) 7/8

(b) 3/8

(c) 1/8

(d) 5/8

(e) 1

**3.** $\lim_{x \to 0^+} (\sin x)^x =$

(a) $\infty$

(b) $e$

(c) $\pi$

(d) 1

(e) 0

**4.** Which of the following is/are True? (I) If $\lim_{x\to c} f(x) = f(c)$, then $f$ is continuous at $c$. (II) If $f(x) = g(x)$ for $x \neq c$ and $f(c) \neq g(c)$, then either $f$ or $g$ must be continuous at $c$. (III) The Intermediate Value Theorem guarantees that $f(a)$ and $f(b)$ differ in sign when a continuous function $f$ has at least one zero on $[a, b]$.

(a) III

(b) II

(c) I and III

(d) I, II and III

(e) I

**5.** If $\sinh x = \dfrac{3}{2}$, then $\tanh x =$

(a) $\dfrac{3\sqrt{13}}{13}$

(b) $\dfrac{3}{13}$

(c) $\dfrac{2\sqrt{13}}{13}$

(d) $\dfrac{\sqrt{13}}{26}$

(e) $3\sqrt{13}$

**6.** The number of vertical asymptotes of the graph of the function

$$f(x) = \frac{x^2 - 9}{x^3 + 3x^2 - x - 3}$$

is

(a) 3

(b) 1

(c) 2

(d) 0

(e) 4

**7.** The value of the constant $a$ that will make the function

$$g(x) = \begin{cases} \dfrac{x^2 - a^2}{x - a}, & x \neq a \\ 8, & x = a \end{cases}$$

continuous everywhere is

(a) there is no such value of $a$

(b) $a = -8$

(c) $a = 8$

(d) $a = -4$

(e) $a = 4$

**8.** The derivative of the function

$$f(x) = \frac{1}{2} \arctan\left(e^{2x}\right)$$

is

(a) $\dfrac{e^{2x}}{1 - e^{4x}}$

(b) $\dfrac{e^{2x}}{1 + e^{4x}}$

(c) $e^{2x} \sec^2\left(e^{2x}\right)$

(d) $\dfrac{1}{2\left(1 + e^{4x}\right)}$

(e) $\dfrac{e^{2x}}{1 - e^{2x}}$

**9.** One statement is False about $f(x) = x - \ln x$

(a) $x = 0$ is a vertical asymptote of $f(x)$

(b) $f(x)$ is concave upwards on its domain

(c) $f(x)$ is increasing over $(0, e)$

(d) The range of $f(x)$ is $[1, \infty)$

(e) The domain of $f(x)$ is $(0, \infty)$

**10.** If $f(x) = \cot x$, then $f''\left(\frac{\pi}{6}\right) =$

(a) $\frac{8\sqrt{3}}{9}$

(b) $-8\sqrt{3}$

(c) $-2\sqrt{3}$

(d) $8\sqrt{3}$

(e) $-4\sqrt{3}$

**11.** $\lim_{x\to-\infty} \frac{x}{\sqrt{x^2+1}} =$

(a) $\infty$

(b) 1

(c) $-\infty$

(d) $-1$

(e) $\frac{1}{2}$

**12.** If the inflection point of $f(x) = 1 + \frac{1}{x} + \frac{1}{x^2}$ is $(a, b)$, then $a + b =$

(a) 0

(b) $-19/11$

(c) $-20/9$

(d) $-3$

(e) $-14/17$

**13.** By using differential the best estimate of $\sqrt{16.5}$ is:

(a) 4.125

(b) 4.050

(c) 4.025

(d) 4.0625

(e) 4.150

**14.** Find the indefinite integral

$$\int \frac{x^4 - 3x^2 + 5}{x^4}\, dx$$

(a) $\frac{x^5}{5} - x^3 + 5x + C$

(b) $x + \frac{3}{x} - \frac{5}{3x^3} + C$

(c) $-3x^2 - \frac{15}{x^2} - 3 + C$

(d) $x - \frac{3}{x} + \frac{5}{x^3} + C$

(e) $x + \frac{1}{x^3} - \frac{1}{x^5} + C$

**15.** The slant asymptote of $f(x) = \frac{2x^3 + x^2 + x - 3}{x^2 + 2x}$ is

(a) $y = -7x + 3$

(b) $y = 2x - 3$

(c) $y = 7x + 3$

(d) $y = 2x + 3$

(e) $y = -2x - 3$

**16.** $\lim_{x \to 2} (2x - \lfloor x \rfloor)$ (where $\lfloor x \rfloor$ is the greatest integer function)

(a) equals 2

(b) equals 1

(c) equals 4

(d) Does not exist

(e) equals 3

**17.** If $y = Ax + B$ is an equation of the normal line to the graph of $x^2 - y^2 = 20$ at the point $(6, 4)$, then $AB =$

(a) 0

(b) $-\frac{16}{3}$

(c) $\frac{3}{2}$

(d) $-\frac{15}{2}$

(e) $\frac{39}{2}$

**18.** The $f(x) = e^{-x} \sin x$ over $[0, 2\pi]$ is decreasing over:

(a) $\left(\frac{\pi}{6}, \frac{3\pi}{2}\right)$

(b) $(0, 2\pi)$

(c) $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$

(d) $\left(0, \frac{5\pi}{4}\right)$

(e) $\left(\frac{\pi}{6}, \frac{5\pi}{4}\right)$

**19.** If $y = (x+1)(x+2)(x+3)(x+4)(x+5)$, then $\frac{d^6y}{dx^6} =$

(a) 5!

(b) 5

(c) 1

(d) 0

(e) 6!

**20.** The derivative of the function $y = x\left(6^{-2x}\right)$ is

(a) $6^{-2x}(1 + 2x \ln 6)$

(b) $-2x6^{-2x} \ln 6$

(c) $6^{-2x}(1 - 2x \ln 6)$

(d) $6^{-2x}\left(\frac{-2x^2}{6} + 1\right)$

(e) $6^{-2x}(1 - 2x)$

# Answer Key

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| Q | Ans | Brief explanation |
|---|---|---|
| 1 | **(b)** | $f = \frac{1}{2}[\ln(x+1) - \ln(x-1)] + \frac{1}{2}\sin 2x$, so $f' = \frac{1}{1-x^2} + \cos 2x$. |
| 2 | **(a)** | $d = 2x^2 - x + 1$, minimum at $x = \frac{1}{4}$: $d = \frac{7}{8}$. |
| 3 | **(d)** | $\ln y = x \ln \sin x \to 0$ (L'Hopital), so the limit is $e^0 = 1$. |
| 4 | **(e)** | Only (I) is true; (II) and (III) have counterexamples. |
| 5 | **(a)** | $\cosh x = \frac{\sqrt{13}}{2}$, so $\tanh x = \frac{3}{\sqrt{13}} = \frac{3\sqrt{13}}{13}$. |
| 6 | **(c)** | Cancel $(x+3)$; remaining denominator $(x-1)(x+1)$ gives 2 asymptotes. |
| 7 | **(e)** | $\lim_{x\to a}(x+a) = 2a = 8$, so $a = 4$. |
| 8 | **(b)** | $f' = \frac{1}{2} \cdot \frac{2e^{2x}}{1+e^{4x}} = \frac{e^{2x}}{1+e^{4x}}$. |
| 9 | **(c)** | $f' = 1 - \frac{1}{x}$: decreasing on $(0,1)$, so 'increasing over $(0,e)$' is false. |
| 10 | **(d)** | $f'' = 2\csc^2 x \cot x$, at $\frac{\pi}{6}$: $2 \cdot 4 \cdot \sqrt{3} = 8\sqrt{3}$. |
| 11 | **(d)** | For $x < 0$, $\sqrt{x^2+1} \approx -x$, so the limit is $-1$. |
| 12 | **(c)** | $f'' = \frac{2x+6}{x^4} = 0 \Rightarrow a = -3$, $b = \frac{7}{9}$, so $a + b = -\frac{20}{9}$. |
| 13 | **(d)** | $\sqrt{16.5} \approx 4 + \frac{1}{8}(0.5) = 4.0625$. |
| 14 | **(b)** | $\int (1 - 3x^{-2} + 5x^{-4})dx = x + \frac{3}{x} - \frac{5}{3x^3} + C$. |
| 15 | **(b)** | Long division: $f = 2x - 3 + \frac{7x-3}{x^2+2x}$, so $y = 2x - 3$. |
| 16 | **(d)** | Left limit 3, right limit 2: the limit does not exist. |
| 17 | **(b)** | $y' = \frac{x}{y} = \frac{3}{2}$; normal $y = -\frac{2}{3}x + 8$, so $AB = -\frac{16}{3}$. |
| 18 | **(c)** | $f' = e^{-x}(\cos x - \sin x) < 0$ on $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$. |
| 19 | **(d)** | Degree-5 polynomial, so the 6th derivative is 0. |
| 20 | **(c)** | Product rule: $y' = 6^{-2x} - 2x \ln 6 \cdot 6^{-2x} = 6^{-2x}(1 - 2x \ln 6)$. |